

GEOLOGICAL SYNTHESIS OF THE ORINOCO OIL BELT, EASTERN VENEZUELA

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The Orinoco Oil Belt, considered to be the largest hydrocarbon accumulation in the world, is located on the southern border of the Eastern Venezuela basin. The stratigraphic column of the area includes rocks of Pre-Cambrian to Recent ages, but more than 90% of the crude oils are found in Miocene sediments.

Three transgressive-regressive cycles with wide regional distributions are recognised in the Tertiary system: Cycle 1 includes the La Pascua, Roblecito and Chaguaramas formations of Oligocene age; and Cycles 2 and 3 include the Oficina, Upper Chaguaramas and Freites formations of Miocene age. Within the cycles, five lithostratigraphic units are defined, and a sedimentological model for Units I and III is consequently established, recognizing north-orientated, wave and tide-influenced, prograding deltas.

Tensional tectonics characterize the area as having two structural provinces, separated by the Hato Viejo fault system: the Eastern Province, characterized by a transgressive sequence overlying the Pre-Cambrian basement; and the Western Province, where the Tertiary unconformably overlies Cretaceous and Paleozoic sediments. Hydrocarbon accumulations in the Orinoco Oil Belt predominantly occur in stratigraphic traps, but are controlled by regional structures.

Four types of crude oils are identified, with density values between 8.5 and 13°API. The volume of oil in-place in the Orinoco Oil Belt totals 187,897 MM † cu. m (1,181 B bbl). The largest accumulations are found in the sands of Unit I.

INTRODUCTION

The Orinoco Oil Belt, the largest hydrocarbon accumulation in the world, is located along the southern border of the Eastern Venezuela basin, and covers an area of some 54,000 sq. km (Fig. 1). Oil exploration in the area began in the 1930s, and continued until 1956 when newly-discovered fields reached a production level of 20,000 b/d. The name of the Orinoco Oil Belt (*Faja Petrolífera del Orinoco*) was given to the area that contained almost exclusively heavy and extra-heavy oil (Galavis and Velarde, 1967; Fiorillo *et al.*, 1981).

Hydrocarbons in the Orinoco Oil Belt predominantly occur in Tertiary clastic sediments, especially in the Miocene, although some reservoirs exist in Cretaceous rocks with very restricted occurrences. Most of the oil produced from the Tertiary is found at depths of less than

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† MM: million (10⁶); B: billion (10⁹).

914 m, although accumulations range in depths from less than 610 m to more than 1,220 m. A new and more extensive exploratory program of the area was established in 1978, when Petróleos de Venezuela assigned to its different affiliated companies the evaluation of four areas from east to west: Cerro Negro (Lagoven S.A.), Hamaca (Meneven S.A.), Zuata (Maraven S.A.) and Machete (Corpoven S.A.) (Fig. 1). The 1979/83 program included the analysis of data from 25,300 km of seismic surveys, 156,500 sq. km of gravimetric, magnetometric and aeromagnetic surveys, 669 exploratory wells and 4,500 m of cores. The exploration program concluded in 1983, and the results were presented in an integrated report prepared for Petróleos de Venezuela, (Fiorillo *et al.*, 1983).

This paper presents a regional synthesis of the stratigraphic and structural framework of the Orinoco Oil Belt (Latreille *et al.*, 1983). The distribution of the sedimentary cycles in the Tertiary, as well as the structural configuration and trap mechanisms are discussed in order to establish their relationship with oil accumulations. A review of the crude oil reserves in-place and of crude quality is also presented (Fiorillo *et al.*, 1983, and Audemard *et al.*, 1984).

STRATIGRAPHY AND SEDIMENTOLOGY

The stratigraphic column in the Orinoco Oil Belt includes rocks of ages ranging from Pre-Cambrian to Recent time (Kiser *et al.*, 1980). Analyses of Tertiary sediments are emphasised in this paper, because they contain most of the hydrocarbon accumulations. The sedimentological framework of the Tertiary was defined from an analysis of the facies distribution, facies relationships and sedimentological models of the sequence, together with some explanations about age determinations in the area.

Pre-Cretaceous

The Pre-Cretaceous in the Orinoco Oil Belt is restricted to the Machete and Zuata areas, and to the NW part of Hamaca; included are rocks of the Hato Viejo, Carrizal and Espino formations, of Paleozoic and Jurassic (?) age, respectively. No hydrocarbon accumulations have been found in these sequences.

Cretaceous

The Cretaceous in the area is represented by sediments of the Temblador Group, and includes the Tigre and Canoa formations, characterized by sands and black shales, and glauconitic limestones, respectively (Dusenbury, 1960).

The Cretaceous extends as a sedimentary east-west unit from the NE margin of Cerro Negro to the western section of Machete. From north to south, the section decreases in thickness until it disappears by pinching out or by erosion. The thickness of the Cretaceous varies from one area to another: maximum thicknesses are located in the NW section of the Machete area, where the sequence reaches a thickness of 914 m. To the east, in Zuata, Hamaca and Cerro Negro, thicknesses can change from 183 m to 30 m. The sequence unconformably overlies the Pre-Cambrian Basement in the Cerro Negro area, and progressively disappears to the south. Towards the Hamaca area, the sediments become thinner and wedge-out to the south. In the Zuata area, they unconformably overlie the Paleozoic, and a Pre-Cretaceous Section (Espino Formation) in Machete, truncated to the south by the Altamira fault (Latreille *et al.*, 1983).

The upper contact of the Cretaceous is defined as a regional angular unconformity, which is normally difficult to recognize because of a sand-sand contact with the Tertiary.

Tertiary

The Tertiary sedimentary column in the Orinoco Oil Belt is represented by three regressive-transgressive cycles, Oligocene and Miocene in age, which have been established by an investigation of palynomorphs (Bolli, 1966; Bolli and Premolisilva, 1973; Dorning 1982) (Figs.

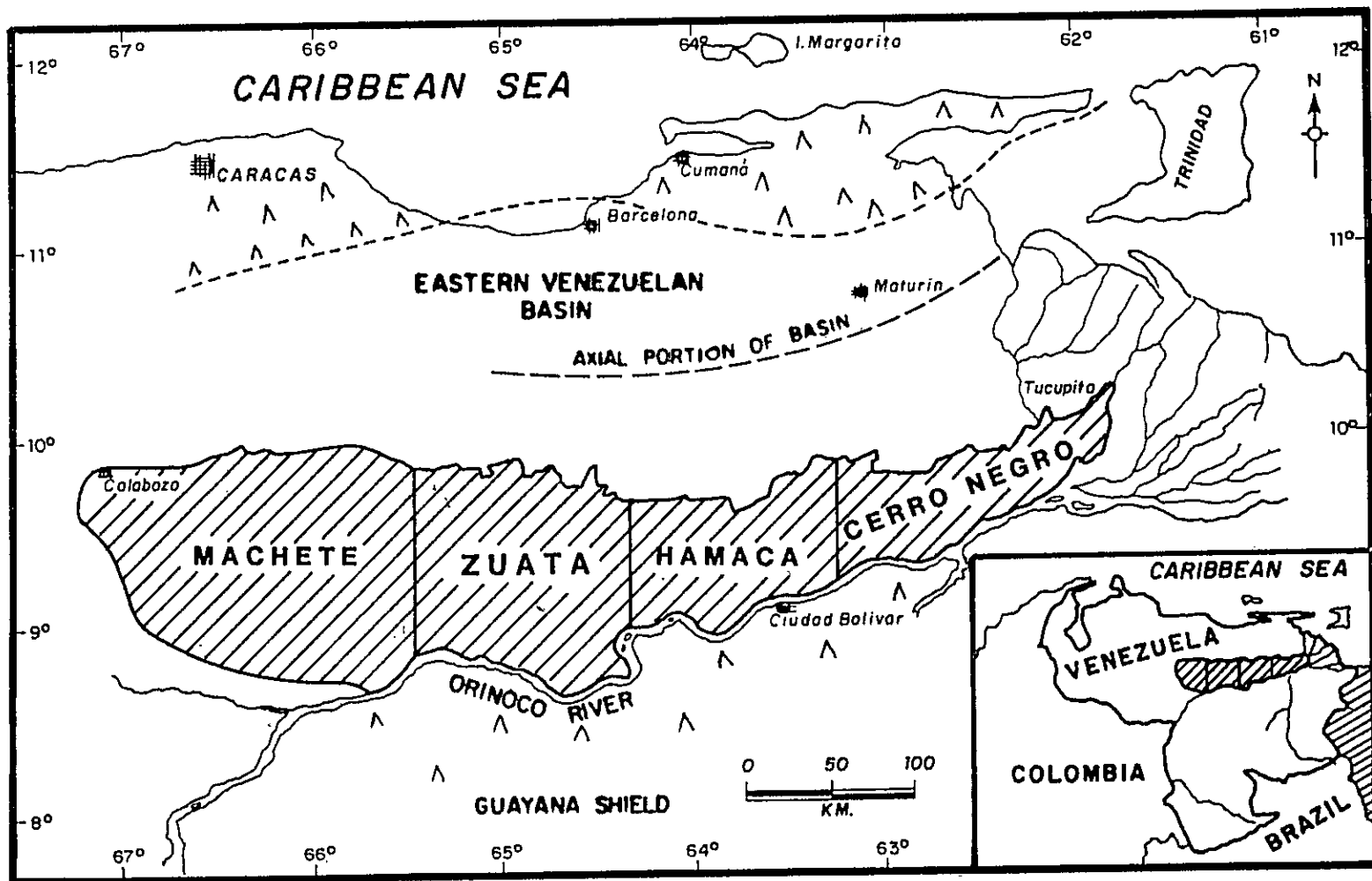


Fig. 1. Location map of the Orinoco Oil Belt.

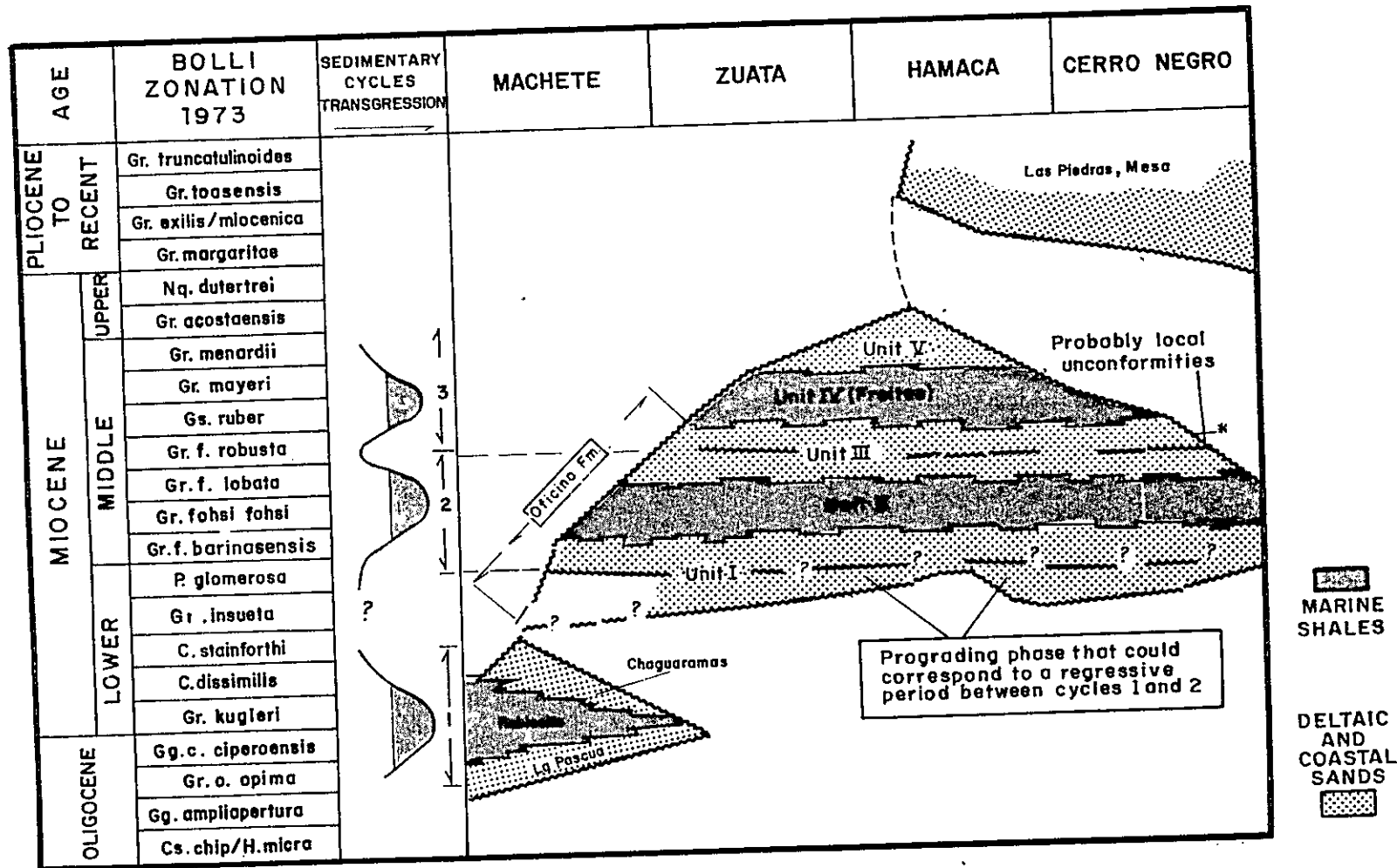


Fig. 2. East-west distribution of the lithostratigraphic nomenclature and age of the sedimentary cycles in the Tertiary of the Orinoco Oil Belt. (After Latreille et al., 1983).

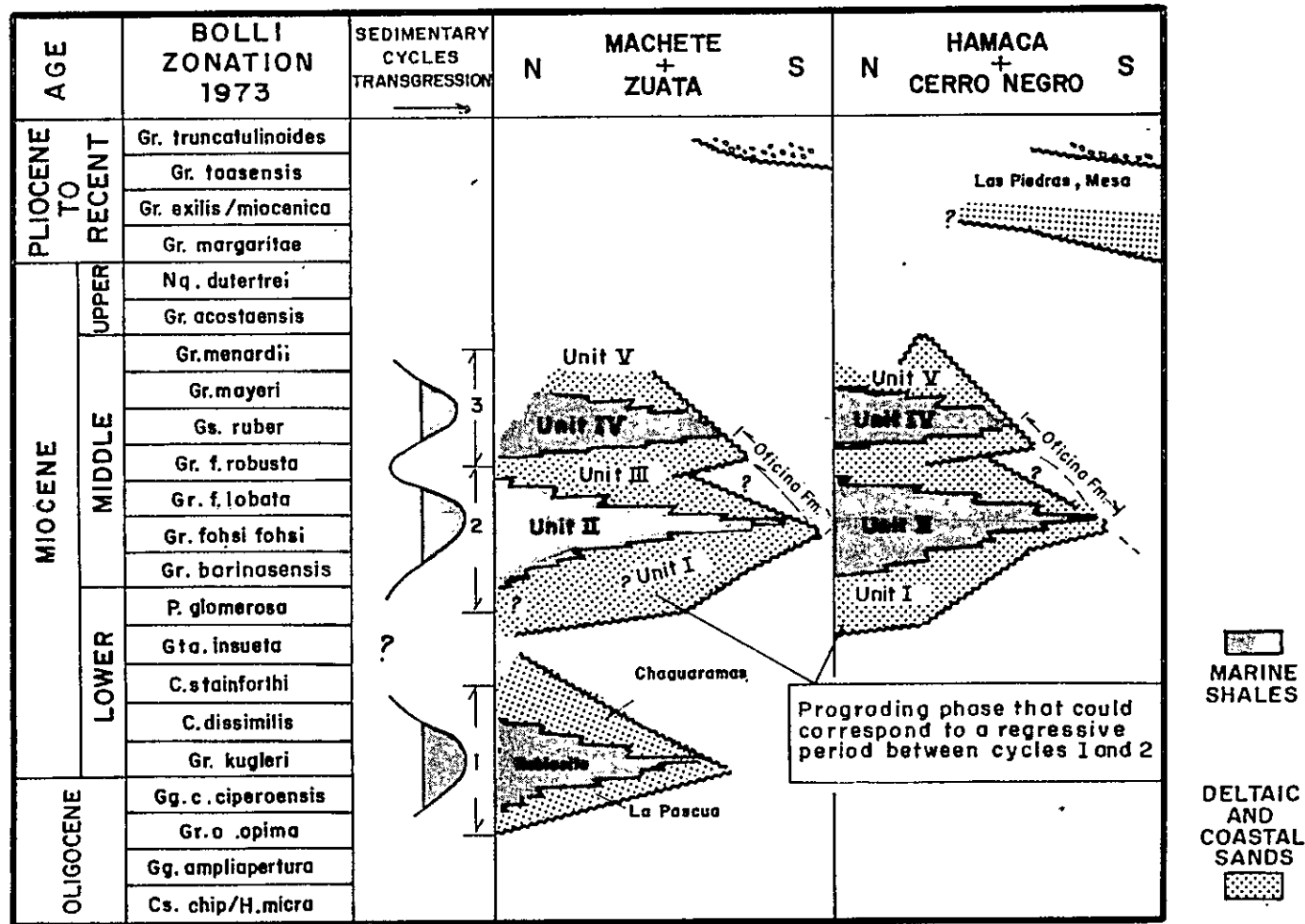


Fig. 3. North-south distribution of the lithostratigraphic nomenclature and age of the sedimentary cycles in the Tertiary of the Orinoco Oil Belt. (After Latreille et al., 1983).

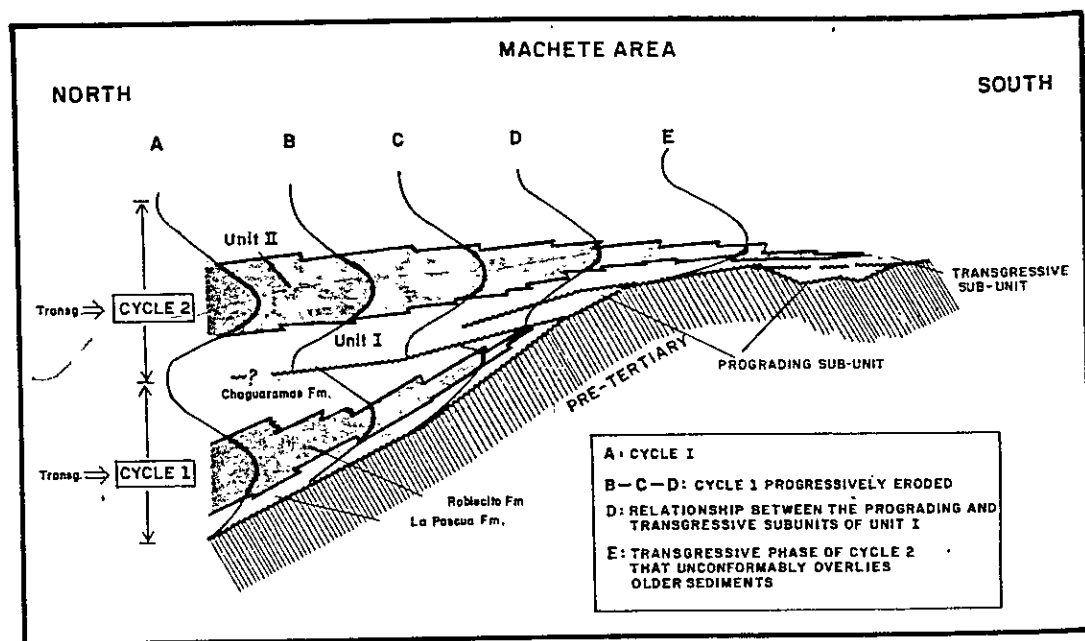


Fig. 4. Schematic north-south section of the Machete area, where the relationship between Cycles 1 and 2 and Unit I is shown. (After Latreille et al., 1983).

2 and 3). The Paleocene and Eocene sections do not occur in this area, but are well represented in the northern part of the Eastern basin (INTEVEP and BEICIP, 1980).

The distribution of the cycles in the area was established from lithostratigraphic correlations, and was based on a horizontal datum that constitutes the peak of the second transgression (Unit II). This datum corresponds in age to the Middle Miocene, determined by the occurrence of *Globorotalia fohsi fohsi* (Bolli and Premolisilva, 1973) (Figs. 2 and 3).

Cycle 1

This is the oldest Tertiary Cycle that can be recognized in the area, and is restricted to the Oligocene (the La Pascua, Roblecito and Chaguaramas formations). It extends over the Machete area and the NW part of Zuata, with thicknesses reaching 914 m to the north.

Cycle 1, also called the regressive-transgressive cycle of the Oligocene, may unconformably overlie either Cretaceous, Pre-Cretaceous or Pre-Cambrian rocks; it appears to have an unconformable contact with Miocene sediments. Sedimentologically, the regressive phase of Cycle 1 displays progressive erosion to the south and east of area (Fig. 4), which may have occurred before the transgressive period of Cycle 2.

The transgressive phase of the cycle is represented by the La Pascua Formation, and is restricted to the central part of the Machete area, with maximum thickness of 122 m and 137 m. This phase is identified as transgressive sheet-sands with a preferential east-west orientation, typically with littoral bar, tidal inlet or shore-face deposits, and with occasional interstratified shales representative of lagoons or estuaries.

The peak of the transgression is represented by the Roblecito Formation, and consists of a shaly sequence that reaches a thickness of 580 m in the northern part of the Machete area. To the east, it laterally grades to sandy sequences, and is rapidly truncated to the south. The lithological characteristics, the facies configurations and the presence of marine fauna indicate a restricted neritic platform environment.

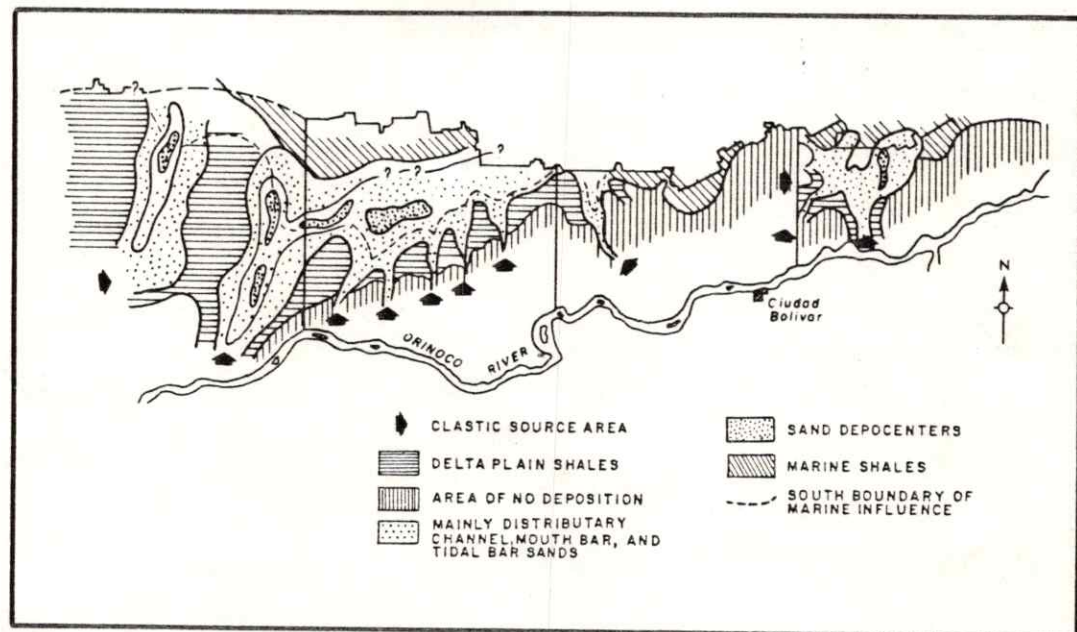


Fig. 5. Delta distribution of the prograding sub-unit of Unit I (see text for explanation).
(After Latreille et al., 1983).

The Chaguaramas Formation represents the upper part of the cycle, and corresponds to a regressive period that occurred after Roblecito time. Its maximum thickness varies between 183 m and 244 m. The unit crops out to the NW, wedges out by erosion to the east and south of the Machete area, and changes laterally to more shaly facies to the north.

Sedimentologically, this facies is interpreted as lightly reworked distributary mouth-bar and lower-shoreface deposits, interstratified with swamp, bay fill or interdistributary bay deposits. Upwards, the regressive sands grade into massive sands which could correspond to stacked, distributary mouth-bar deposits. In this cycle, several hydrocarbon accumulations have been reported (Corpovent, 1981).

Cycles 2 and 3

Cycles 2 and 3 are of Miocene age, and represent the most important cycles of the Tertiary on account of their hydrocarbon accumulations. They have been subdivided into five informal lithostratigraphic units which can be regionally correlated, and represent alternating thicker intervals of sands and shales.

Cycle 2 begins with a predominantly sandy basal interval (Unit I) that grades upwards to a shaly interval (Unit II). The shales represent the maximum peak of the transgression of Cycle 2. Up-section, the shales grade into regressive sands which indicate the beginning of Cycle 3 and constitute Unit III. Because the boundary of Cycles 2 and 3 is defined in Unit III, both will be discussed together. Units I to II correspond to the Oficina Formation. In the upper part of the unit, the sands grade into shales of Unit IV, the Freites Formation, and represent the maximum peak of Cycle 3, overlain by the regressive sands of Unit V (Figs. 2 and 3). A brief sedimentological description of Units I to V follows:

Unit I

This unit is predominantly sandy, and extends throughout the whole area with a maximum thickness of 305 m to the north of the Zuata and Machete areas. To the south, it continuously

thins until disappearing. The Unit can be subdivided into two subunits: a prograding and a regressive subunit.

The prograding subunit consists of massive sands which represent the deposition of several delta complexes. These deltas seem to have been fed by rivers flowing from the south, and grade to the north into more marine facies. The paleofacies map (Fig. 5, p. 141) shows an area of non-deposition that was cut by these north-oriented rivers, and later on, covered by transgressive beach and barrier complexes. To the north, and connected with areas of non-deposition, some delta plain deposits occur, dissected by northerly-trending narrow areas that represent thick sandy deposits (> 50% sand), with a horizontal distribution that becomes wider to the north. These narrow extensions correspond to distributary channels which extend upwards to form the sand depocenters. The northern limit of these delta plain deposits is indicated by a dashed line (Fig. 5) that separates these deposits from the sub-aqueous delta front. The shale content also increases to the north, where sands grade to offshore neritic sediments.

The transgressive sub-unit is well distributed in the Cerro Negro and Hamaca areas. The sedimentary system has an east-west direction, parallel to the southern border of the basin. This orientation contrasts with the lower system where the main directions are perpendicular to the area. Based on correlations and core analysis (Isea, 1981, 1982; Lambertini, 1980 a and b; Lambertini, 1982; and Latreille *et al.*, 1983), the sub-unit can be interpreted as a beach-barrier island system, including transgressive sheet sands associated with tidal inlet, shoreface and washover sands, similar to those of the La Pascua Formation.

The main hydrocarbon accumulations of the Orinoco Oil Belt are located in the regressive sub-unit of Unit I.

Unit II

Unit II represents the maximum peak of Cycle 2, and consists of a predominant shaly sequence which constitutes the main seal for the hydrocarbon accumulations existing in the underlying Unit I. It extends from the northern part of Cerro Negro and continues up to the northern part of Hamaca and Zuata. To the south, it extends up to the Orinoco River, where it disappears by wedging out. Sedimentologically, this shaly sequence has occasional interstratifications of sand and siltstones, and in conjunction with the existence of the marine fauna, appears to indicate a restricted marine environment.

Unit III

Unit III is predominantly sandy, and indicates the maximum phase of the Middle Miocene regression. The unit conformably overlies Unit II, with only local unconformities (Fig. 2), and is also conformably overlain by Unit IV. Towards the Cerro Negro and Hamaca areas, its thickness reaches 548 m, and decreases to the south due to erosion. The unit crops out over the rest of the areas but to the west and NW of Machete, it is completely eroded (Figs. 2 and 6). In Zuata, the unit can be as much as 396-m thick, but in the northern portions it is reduced to 152 m in thickness and it grades into shaly sequences.

Facies association suggests that Unit III represents an extensive sheet-sand which may correspond to a large, prograding delta. These sands have an east-west orientation, and run parallel to former beach lines in a beach-barrier island system with characteristic tidal inlet, washover or barrier deposits. Some lagoon and bay-fill shales occur interstratified with these sands. Pollen data support a lower delta-plain environment for these deposits. Up-section, these sand bodies decrease in lateral extent and grade to a massive sand that appears to represent the delta depocenter. This massive sand can be followed for more than 80 kms, reaches a maximum thickness of 137-177 m in the northern part of Hamaca (Fig. 6), and gradually changes to sheet-sands to the east and west of this area. To the south, the sand body loses its massive character and grades into distributary channel deposits, interstratified with stacked point bars (upper delta plain deposits).

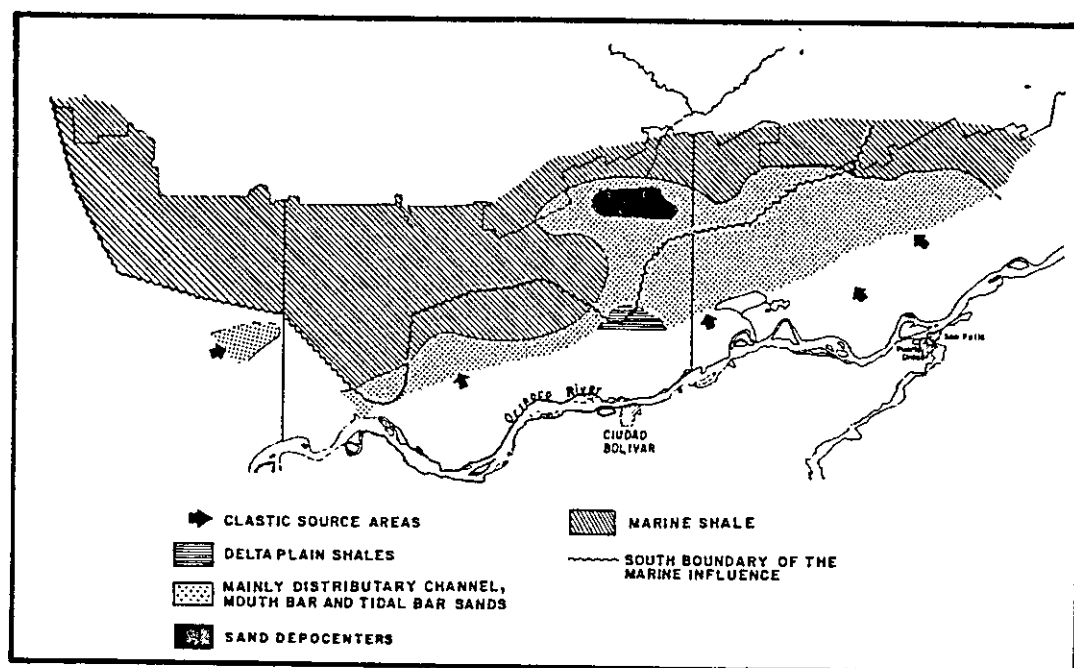


Fig. 6. Delta distribution of Unit III. (After Latreille *et al.*, 1983).

Overlying these massive sands, the sequence starts to be transgressive (Cycle 3), with several intervals of marine fauna recognizable.

Unit IV

This Unit represents the second Miocene transgression. The sequence is predominantly shaly and corresponds to the Freites Formation. Its thickness gradually increases in the northern part of the Hamaca area where it is 183-m thick. In the south, the transgressive sequence disappears by sedimentary thinning but not by erosion.

Because of its eminently shaly character and the presence of marine fauna, this unit is considered to be of neritic origin.

Unit V

Unit V conformably overlies Unit IV, and constitutes the regressive phase of Cycle 3. The Unit is present only in the northern part of Hamaca and in the NW part of Cerro Negro and Zuata where erosion did not affect it. Marine fossils and glauconite in some samples appear to indicate its marine character. However, sedimentary processes have not yet been well established.

Post-Miocene deposits

The post-Miocene deposits in the Orinoco Oil Belt correspond to a predominantly sandy unit with continental character (Las Piedras Formation), which unconformably overlies Miocene sediments in Cerro Negro and Hamaca. Its extension is restricted to these two areas, and its age is defined as Upper Miocene-Pliocene (González de Juana *et al.*, 1980).

In summary, the Tertiary in the Orinoco Oil Belt is represented by a very short sequence that includes Oligocene and Miocene sediments, deposited in the coastal-influenced continental province of the Eastern Venezuela basin, as is evidenced by the interaction of marine and continental influences. During the transgressive phases, several strong marine incursions

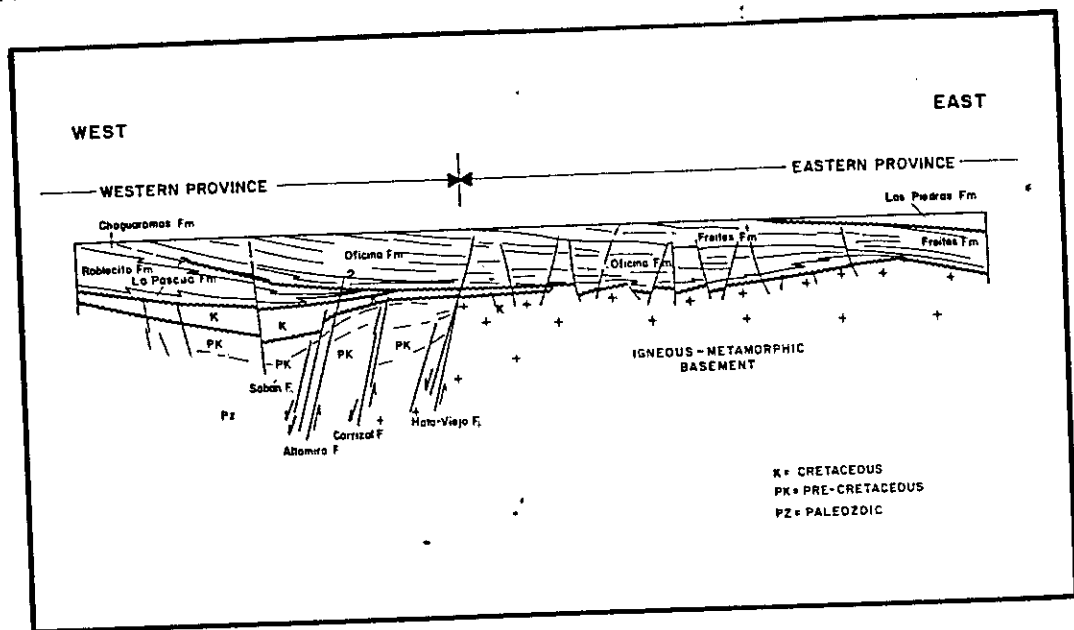


Fig. 7. Schematic structural configuration of the Orinoco Oil Belt. (After Latreille et al., 1983).

occurred, while the continental influence became more evident in the north during the regressive phases. As a consequence, three transgressive-regressive cycles are identified in the area; these are most clearly expressed in the north, being truncated to the south. The facies distribution of the different units identified in the cycles indicates the existence of large tide-wave dominated deltas (Galloway, 1975), associated with regressive phases and recognized in Units I and III, where the thickest sandy bodies are developed.

STRUCTURE AND TECTONICS

The Orinoco Oil Belt consists of a prism of Tertiary sediments wedging to the south, which unconformably overlies the Cretaceous, the Paleozoic or the Pre-Cambrian basement.

The structural character of the area is represented by tensional tectonics, established by the integration of regional seismic maps, and based on structural sections that include the lithostratigraphic units already mentioned. Two different provinces can be recognized, separated by the Hato Viejo fault system (Fig. 7): the Eastern Province (the Cerro Negro and Hamaca areas), where Tertiary sediments overlie the Pre-Cambrian Basement, except for a narrow rim where the Tertiary transgresses over Cretaceous rocks; and the Western Province (the Zuata and Machete areas), where Tertiary sediments overlie Cretaceous, Jurassic and Paleozoic deposits, the last of which are preserved in deep structural depressions.

Regionally, the dynamics of the area correspond to fault tectonics, characterized by rigid blocks, without evidence of folding. The average vertical displacement does not exceed 61 m and the faults are mainly of a tensional type (normal), regardless of whether they are parallel or normal to the main trend (Fig. 7). Similarly, some horizontal displacements between blocks, related to strike-slip faults, also occur, but they have not yet been studied in detail. Some secondary compressive effects have also identified, and may be the result of the same dynamics.

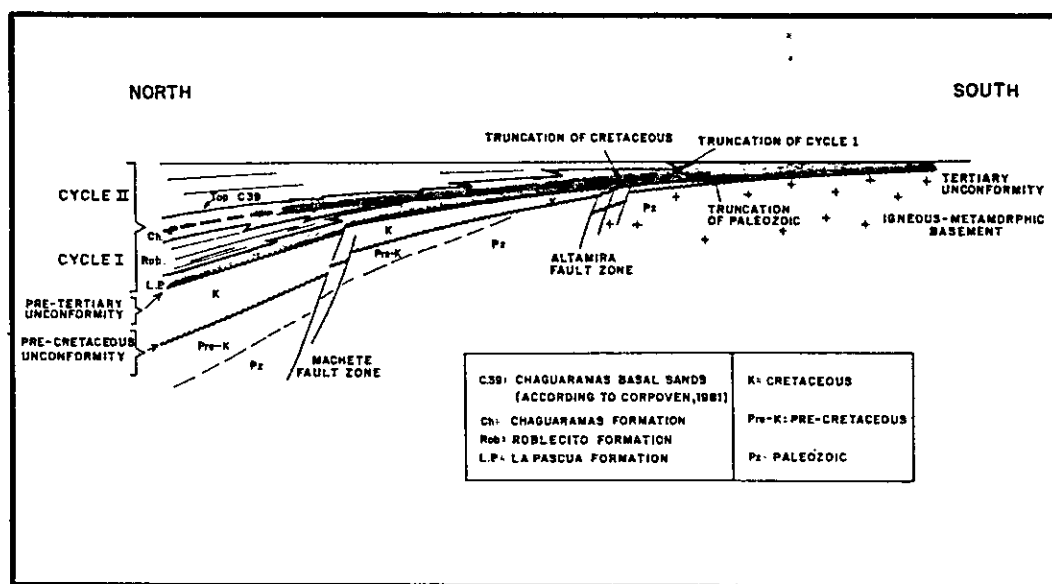


Fig. 8. Relative distribution of unconformities and faults in the Machete area. (After Latreille et al., 1983).

Three preferential major tectonic trends are recognized in the Eastern Province, where a number of faults affect the Basement and the top of the Oficina Formation: an east-west trend, along the hinge-zone of north Hamaca and Cerro Negro; a $N60^{\circ}-70^{\circ}E$ trend, parallel to the orientation of the Guayana Shield, south of the Orinoco river; and a $N30^{\circ}-45^{\circ}W$ trend, a prominent orientation which reflects the Pre-Tertiary paleotopography, traverse to the length of the belt, and with depressions that were filled with fluvio-deltaic deposits at the beginning of the Tertiary.

In the Western Province, the dominant direction of the major faults varies from east-west to NE-SW. The NE-trending faults are restricted to the Machete area, where the Altamira fault system determines the structural boundary between the southern and northern areas (Fig. 8). The structural configuration of the Machete area also permits the identification of a NE-trending "high" (the Monasterio Arch), which coincides with the top of the Chaguaramas Formation.

The structural control of hydrocarbon distribution in the area is secondary to the stratigraphic control of the accumulations, which appear as an enormous, heavy-oil "seal" migrated and trapped in the Tertiary sands (Unit I). Structural control is observed on a regional scale in some structural highs (Machete and Monasterio) and in the hinge zones (Hamaca-Cerro Negro). However, local and detailed analyses indicate that oil distribution is not necessarily controlled by the presence of faults. Furthermore, it can be located in either the upthrown or downthrown sides of these faults.

CRUDE OIL DISTRIBUTION AND CHARACTERIZATION

Volumetric estimates of "in-place" oil reserves in the area were obtained from the evaluation of the 662 drilled wells, and from the integration of geophysical and geological data. The total

volume of oil in-place in the Orinoco Oil Belt amounts to 187,879 MM cu. m (1,181 B brl) and its distribution by each operational area is:

Cerro Negro	Hamaca	Zuata	Machete
33,826 MM cu. m 213,000 MM brl	34,022 MM cu. m 214,000 MM brl	79,491 MM cu. m 499,000 MM brl	40,540 MM cu. m 255,000 MM brl

According to its distribution in the stratigraphic sequence, 79% of the oil is present in the sands of Unit I; 4% occurs in the sand-shale interstratifications of Unit II; 8% is present in the upper sands of Unit III; and 9% occurs in Cretaceous and Oligocene sediments.

Based on a study of the physical and chemical properties of 288 crude oil samples (Audemard *et al.*, 1984; Taheri and Audemard, 1984), including density, sulphur (S), Vanadium (V) and asphaltene content, and viscosity at different temperatures, four groups of crude oils were identified in the Orinoco Oil Belt, with the following characteristics:

Crude Oil Type	Density °API	Viscosity at 100°C (cSt)	% S	V (ppm)
A	13	60	1.60	250
B	10-13	60-230	1.60-3.24	250-340
C	8.5-10	200-300	3.24-3.80	340-450
D	8.5	300	3.80	450

Type A crude oils occur in the NE part of Hamaca and Zuata, to the south of Hamaca, and in the eastern part of Cerro Negro. B quality crude oils are distributed to the north of the Zuata and Hamaca areas, and in some local areas of Cerro Negro. "Extra-heavy" crude oils of qualities C and D are found in the south of the area, from Machete to Cerro Negro; in the latter area, type D crude oils occur to the north and cover the greatest part of the Orinoco Belt.

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